

GPU_CVT Paper Review – Priority Improvements

High-Impact Issues (Critical)

These issues must be addressed before submission.

- 1 Separate KNN backend improvement from freeze policy via ablation (3 variants: brute-force, bitonic only, bitonic+freeze).
- 2 Add comparisons against alternative acceleration strategies (early stopping, approximate CVT, GPU baselines).
- 3 Perform sensitivity analysis on NV thresholds, streak lengths, and displacement threshold.

Medium-Impact Issues

These strengthen the paper significantly.

- 1 Add runtime breakdown (KNN vs centroid vs projection).
- 2 Plot frozen fraction over iterations.
- 3 Test generalization (weighted CVT or different K values).
- 4 Measure overhead of freeze logic and memory cost.

Low-Impact Improvements

Polish and positioning.

- 1 Reframe as adaptive computation / skewness-aware parallelism.
- 2 Expand related work on dynamic workload pruning.
- 3 Improve figures (pipeline clarity, workload shrink visualization).

Strengths of the Paper

- 1 Strong causal analysis of convergence failure.
- 2 Clean dual-gate design.
- 3 Large-scale evaluation.
- 4 Clear performance narrative.